

SQL Practice Workbook (Extended)

■ Sample Schema

Customers

customer_id	first_name	last_name	email
1	Alice	Smith	alice@mail.com
2	Bob	Johnson	bob@mail.com
3	Charlie	Brown	charlie@mail.com
4	Diana	White	diana@mail.com

Orders

order_id	customer_id	order_date	total_amount	status
101	1	2025-09-05	200.00	C
102	2	2025-09-04	150.00	P
103	1	2025-09-07	300.00	C
104	3	2025-09-08	120.00	R

OrderItems

order_item_id	order_id	product_id	quantity	price
1	101	201	2	50.00
2	101	202	1	100.00
3	102	202	3	50.00
4	103	203	5	60.00
5	104	201	2	60.00

Products

product_id	product_name	category	price
201	Laptop Bag	Accessories	50.00
202	Mouse	Electronics	50.00
203	Keyboard	Electronics	60.00

1■■■ Total Orders & Spend per Customer

Query:

```
SELECT c.first_name || ' ' || c.last_name AS customer_name, COUNT(o.order_id) AS
total_orders, SUM(o.total_amount) AS total_spent FROM Customers c JOIN Orders o ON
c.customer_id = o.customer_id GROUP BY c.first_name, c.last_name ORDER BY total_spent DESC;
```

Expected Output:

customer_name	total_orders	total_spent
Alice Smith	2	500.00
Bob Johnson	1	150.00
Charlie Brown	1	120.00

2■■■ Top Customers by Quantity Purchased

Query:

```
SELECT c.first_name || ' ' || c.last_name AS customer_name, SUM(oi.quantity) AS
total_quantity, SUM(oi.quantity * oi.price) AS total_spent FROM Customers c JOIN Orders o ON
c.customer_id = o.customer_id JOIN OrderItems oi ON o.order_id = oi.order_id GROUP BY
c.first_name, c.last_name ORDER BY total_quantity DESC LIMIT 5;
```

Expected Output:

customer_name	total_quantity	total_spent
Alice Smith	8	500.00
Bob Johnson	3	150.00
Charlie Brown	2	120.00

3■■■ Most Popular Product Category

Query:

```
SELECT p.category, SUM(oi.quantity * oi.price) AS total_sales FROM OrderItems oi JOIN
Products p ON oi.product_id = p.product_id GROUP BY p.category ORDER BY total_sales DESC;
```

Expected Output:

category	total_sales
Electronics	450.00
Accessories	170.00

4■■■ Customers with No Orders

Query:

```
SELECT c.customer_id, c.first_name || ' ' || c.last_name AS customer_name FROM Customers c
LEFT JOIN Orders o ON c.customer_id = o.customer_id WHERE o.order_id IS NULL;
```

Expected Output:

customer_id	customer_name
4	Diana White

5■■■ Latest Order Per Customer

Query:

```
SELECT c.first_name || ' ' || c.last_name AS customer_name, MAX(o.order_date) AS latest_order_date, o.total_amount FROM Customers c JOIN Orders o ON c.customer_id = o.customer_id GROUP BY c.first_name, c.last_name, o.total_amount ORDER BY latest_order_date DESC;
```

Expected Output:

customer_name	latest_order_date	total_amount
Charlie Brown	2025-09-08	120.00
Alice Smith	2025-09-07	300.00
Bob Johnson	2025-09-04	150.00

6■■■ Orders with Product Details

Query:

```
SELECT o.order_id, p.product_name, oi.quantity, (oi.quantity * oi.price) AS line_total FROM Orders o JOIN OrderItems oi ON o.order_id = oi.order_id JOIN Products p ON oi.product_id = p.product_id ORDER BY o.order_id;
```

Expected Output:

order_id	product_name	quantity	line_total
101	Laptop Bag	2	100.00
101	Mouse	1	100.00
102	Mouse	3	150.00
103	Keyboard	5	300.00
104	Laptop Bag	2	120.00

7■■■ Total Sales by Customer & Category

Query:

```
SELECT c.first_name || ' ' || c.last_name AS customer_name, p.category, SUM(oi.quantity * oi.price) AS total_spent FROM Customers c JOIN Orders o ON c.customer_id = o.customer_id JOIN OrderItems oi ON o.order_id = oi.order_id JOIN Products p ON oi.product_id = p.product_id GROUP BY c.first_name, c.last_name, p.category ORDER BY total_spent DESC;
```

Expected Output:

customer_name	category	total_spent
Alice Smith	Electronics	300.00
Alice Smith	Accessories	200.00
Bob Johnson	Electronics	150.00
Charlie Brown	Accessories	120.00

8■■■ Customers with Orders Above Threshold

Query:

```
SELECT c.first_name || ' ' || c.last_name AS customer_name, o.order_id, o.total_amount FROM
Customers c JOIN Orders o ON c.customer_id = o.customer_id WHERE o.total_amount > 200 ORDER
BY o.total_amount DESC;
```

Expected Output:

customer_name	order_id	total_amount
Alice Smith	103	300.00

9■■■ Product-Wise Customer Purchases

Query:

```
SELECT p.product_name, c.first_name || ' ' || c.last_name AS customer_name, SUM(oi.quantity)
AS total_quantity FROM Customers c JOIN Orders o ON c.customer_id = o.customer_id JOIN
OrderItems oi ON o.order_id = oi.order_id JOIN Products p ON oi.product_id = p.product_id
GROUP BY p.product_name, customer_name ORDER BY p.product_name, total_quantity DESC;
```

Expected Output:

product_name	customer_name	total_quantity
Laptop Bag	Alice Smith	2
Laptop Bag	Charlie Brown	2
Mouse	Bob Johnson	3
Mouse	Alice Smith	1
Keyboard	Alice Smith	5